Exercise: Show that

$$nI(\theta) = E\Big[-\frac{\partial^2}{\partial\theta^2}\underbrace{\log L(\theta)}_{\ell(\theta)}\Big]$$

Still assuming $X_1, X_2, \dots, X_n$ iid f_X

$$nI(\theta) = n\,E\left(-\frac{\partial^2 \log f(X;\theta)}{\partial\theta^2}\right) \qquad X \sim f_X$$

$$= \sum_{i=1}^{n} E\left(-\frac{\partial^2 \log f(X_i;\theta)}{\partial\theta^2}\right)$$

$$= E\left(-\sum_{i=1}^{n} \frac{\partial^2 \log f(X_i;\theta)}{\partial\theta^2}\right)$$

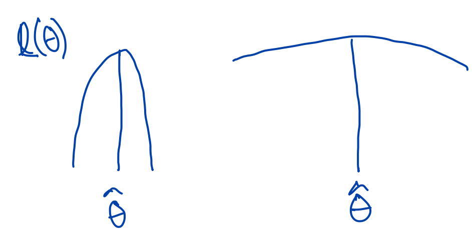
$$= E\left(-\frac{\partial^2 \sum \log f(X_i;\theta)}{\partial\theta^2}\right)$$

$$= E\left(-\frac{\partial^2 \log L(\theta)}{\partial\theta^2}\right)$$

The previous result shows that the variance of the MLE is approximately equal to the reciprocal of the expected value of the negative second derivative of the log likelihood at its peak. $V(\hat\theta) \approx \frac{1}{nI(\theta_0)} \approx \frac{1}{nI(\hat\theta)}$

And, "the negative of the second derivative" is the amount of curvature at the peak of the likelihood.

If the log likelihood has a "sharp" peak, then $nI(\theta)$ will be large, and the variance of $\hat\theta$ will be small.



larger $nI(\hat\theta)$ smaller $V(\hat\theta)$

smaller $nI(\hat\theta)$ larger $V(\hat\theta)$

Exercise: Assume $X_1, X_2, \ldots, X_n$ are i.i.d. $N(\mu, \sigma^2)$, where σ^2 is known. We know that $\overline{X}$ is the MLE for μ. Derive the Fisher Information for μ, and use it to approximate the standard error and distribution of $\overline{X}$.

$$\ell(\mu) = \frac{-\sum (X_i - \mu)^2}{2\sigma^2} + \text{constants that do not depend on } \mu$$

$$\frac{\partial \ell(\mu)}{\partial \mu} = \frac{\sum (X_i - \mu)}{\sigma^2} = \frac{\sum X_i - n\mu}{\sigma^2}$$

$$\frac{\partial^2 \ell(\mu)}{\partial \mu^2} = -\frac{n}{\sigma^2}$$

$$nI(\mu) = E\left(-\frac{\partial^2 \ell(\mu)}{\partial \mu^2}\right) = E\left(-\left(-\frac{n}{\sigma^2}\right)\right) = \frac{n}{\sigma^2}$$

So, the asymptotic normality result says that $\bar{X}$ is approx normal with mean μ_0 and variance $\frac{1}{nI(\mu_0)} = \frac{\sigma^2}{n}$

We already knew that $\bar{X}$ is exactly $N(\mu_0, \sigma^2/n)$

Exercise: Assume that $X_1, X_2, \ldots, X_n$ are i.i.d. Exponential(λ_0). Derive the Fisher Information for λ and the approximate distribution of its MLE.

$\hat{\lambda}_{MLE} = 1/\bar{x}$ $\quad f(x;\lambda) = \lambda e^{-\lambda x}, \quad x > 0$

$$\log f(x;\lambda) = \log \lambda - \lambda x, \quad x > 0$$

$$\frac{\partial \log f(x;\lambda)}{\partial \lambda} = \frac{1}{\lambda} - x$$

$$\frac{\partial^2 \log f(x;\lambda)}{\partial \lambda^2} = -\frac{1}{\lambda^2}$$

So, $I(\lambda) = E\left[-\frac{\partial^2 \log f(x;\lambda)}{\partial \lambda^2}\right] = E\left[\frac{1}{\lambda^2}\right] = \frac{1}{\lambda^2}$

i.e. $nI(\lambda) = \frac{n}{\lambda^2}$, $V(\hat{\lambda}_{MLE}) \approx \frac{1}{nI(\lambda_0)} = \frac{\lambda_0^2}{n} \approx \frac{\hat{\lambda}^2}{n} = \frac{1}{n\bar{x}^2}$

Can claim that $\hat{\lambda}_{MLE}$ is approx. normal with variance $\frac{1}{n\bar{x}^2}$

or SE of $\frac{1}{\sqrt{n}\,\bar{x}}$

Exercise: Assume that $X_1, X_2, \ldots, X_n$ are i.i.d. Poisson(λ). Derive the Fisher Information for λ and the approximate distribution of its MLE.

Efficiency of the MLE

It is possible to show that the lower bound on the variance of any unbiased estimator is $1/nI(\theta_0)$. This is called the Cramer-Rao lower bound.

Since the MLE is asymptotically unbiased, we know that we are achieving the Cramer-Rao lower bound, at least asymptotically.

Hence, we are minimizing the MSE (asymptotically). This is objectively the best rationale for utilizing maximum likelihood estimators.

The estimator is "efficient"

The Delta Method

Often, our objective is not to estimate θ, but instead some function of θ, call it $g(\theta)$. For example, often want to estimate the probability of some event A. This probability is not a parameter, but it is a function of θ.

From the invariance property we know that the MLE of $g(\theta)$ is $g(\widehat{\theta})$. But, we also need to calculate a SE or construct a confidence interval for this estimate.

One approach would be to "reparameterize" the model in terms of $g(\theta)$, and then find $I(g(\theta))$, and then use $1/I(g(\theta))$ as the variance, etc. Fortunately, there is an easier way…

The Delta Method

A **formal statement** of the **Delta Method** is as follows:

> Assume that $Y_1, Y_2, \ldots$ are such that
>
> $$\frac{Y_n - \mu}{\sigma_n} \xrightarrow{D} Z$$
>
> where $Z \sim N(0,1)$, μ is a constant, and σ_n is such that $\sigma_n > 0$ and $\sigma_n \downarrow 0$ as $n \to \infty$. Then, assuming $g(\cdot)$ is a function that is differentiable at μ and $g'(\mu) \neq 0$, it holds that
>
> $$\frac{g(Y_n) - g(\mu)}{|g'(\mu)|\sigma_n} \xrightarrow{D} Z.$$

A more **heuristic**, but useful, version is as follows:

> $\overline{X}_n$ $\quad$ σ^2/n
>
> If Y_n is approximately $N(\mu, \sigma_n^2)$ and $\sigma_n^2 \downarrow 0$ as $n \to \infty$, then $g(Y_n)$ is approximately
>
> $$N\big(g(\mu), (g'(\mu))^2\sigma_n^2\big)$$
>
> assuming that $g(\cdot)$ is a function that is differentiable at μ and $g'(\mu) \neq 0$.

This is often (but not always) applied with $Y_n = \overline{X}_n$ and $\sigma_n^2 = \sigma^2/n$ where $\sigma^2 > 0$.

Exercise: Suppose that $X_1, X_2, \ldots, X_{50}$ are i.i.d. random variables with the Exponential($\lambda = 2$) distribution. One may hope that $1/\overline{X}$ is a good estimator of λ (imagining a case where λ is not known). What is the probability that $1/\overline{X}$ is within 0.1 of λ?

$\mu = E(X_i) = 1/\lambda = 1/2$

$\sigma^2 = V(X_i) = 1/\lambda^2 = 1/4$

By CLT, $\bar{X} \approx N(\mu, \sigma^2/n) = N(1/\lambda, 1/n\lambda^2)$

What is the (approx.) dist. of $1/\bar{X}$?

Use the Δ-method with $g(\bar{X}) = 1/\bar{X}$

$g(x) = 1/x$

So, $g'(x) = -1/x^2$, $g'(\mu) = -\lambda^2 \neq 0$

By the Δ-method, $g(\bar{X}) = 1/\bar{X}$ is approx. Normal with mean $g(\mu) = \frac{1}{1/\lambda} = \lambda$ and variance

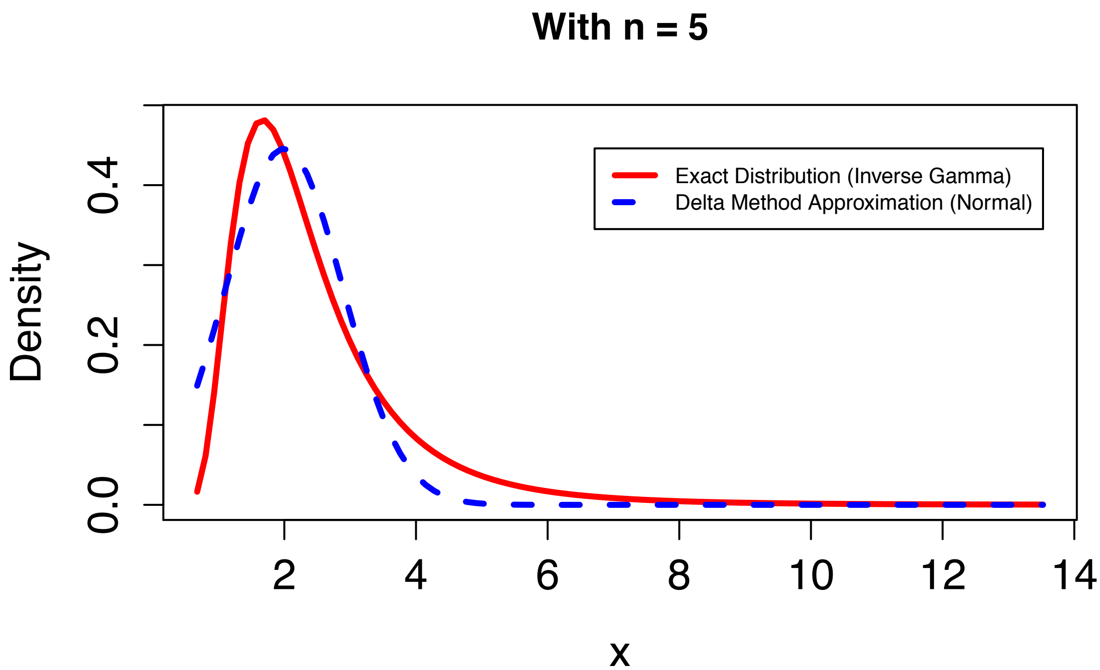
$$(g'(\mu))^2 \frac{\sigma^2}{n} = \lambda^4 \left(\frac{1}{\lambda^2 n}\right) = \frac{\lambda^2}{n}$$

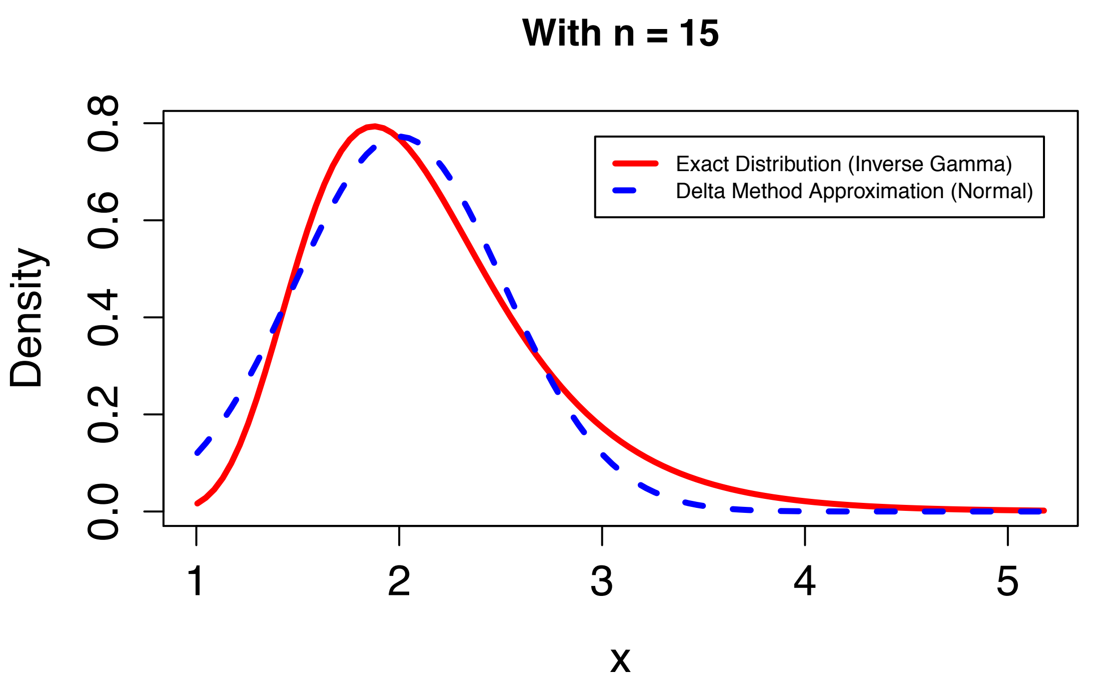
Can claim that by the Δ-method,

$$1/\bar{X} \approx N\left(\lambda, \lambda^2/n\right)$$

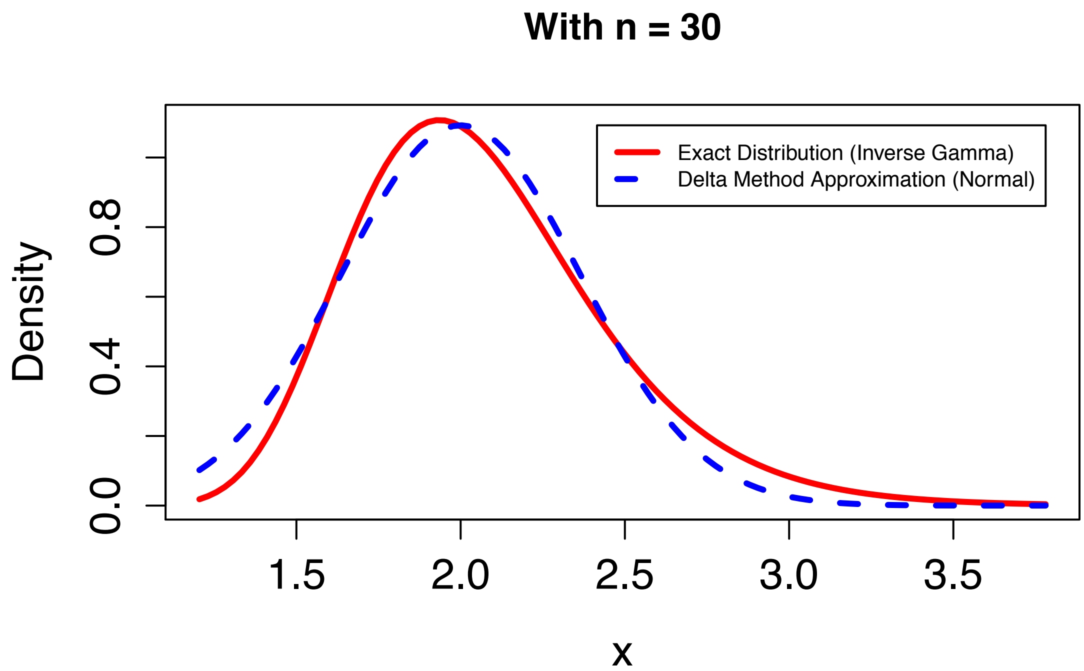
In fact, $1/\bar{X} \sim \text{Inverse Gamma}(n, n\lambda)$

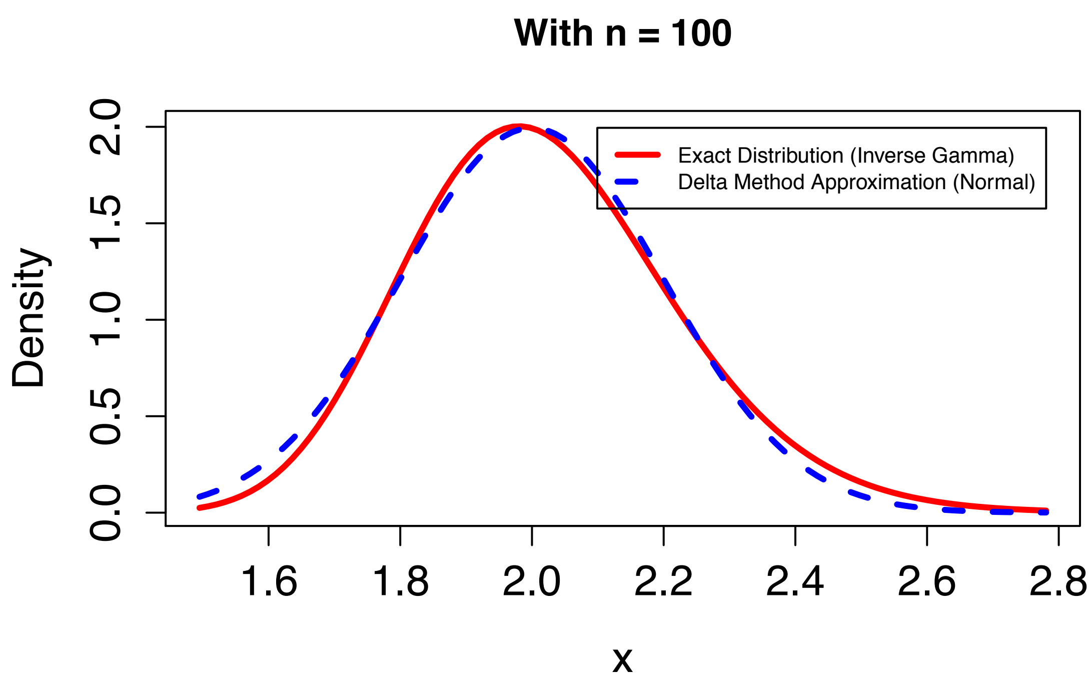
Plug in $\lambda = 2$ and compare these


With n = 5
Exact Distribution (Inverse Gamma)
Delta Method Approximation (Normal)
Density
0.0
0.2
0.4
2
4
6
8
10
12
14
x


With n = 15
Exact Distribution (Inverse Gamma)
Delta Method Approximation (Normal)
Density
0.0
0.2
0.4
0.6
0.8
1
2
3
4
5
x

1


With n = 30
Exact Distribution (Inverse Gamma)
Delta Method Approximation (Normal)
Density
0.0
0.4
0.8
1.5
2.0
2.5
3.0
3.5
x


With n = 100
Exact Distribution (Inverse Gamma)
Delta Method Approximation (Normal)
Density
0.0
0.5
1.0
1.5
2.0
1.6
1.8
2.0
2.2
2.4
2.6
2.8
x

Recall the Delta Method:

$\hat{\theta}$ θ_0

> If Y_n is approximately $N(\mu, \sigma_n^2)$ and $\sigma_n^2 \to 0$ as n increases, then $g(Y_n)$ is approximately $N(g(\mu), (g'(\mu))^2\sigma_n^2)$, assuming that $g'(\mu)$ exists and is not zero.

Here we will use this result with the MLE for θ in place of Y_n.

Then, $\mu = \theta_0$ and $\sigma_n^2 = 1/nI(\theta_0)$.

We can conclude that $g(\hat{\theta})$ is approximately

$$N\left(g(\theta_0),\ g'(\theta_0)^2\left(\frac{1}{nI(\theta_0)}\right)\right)$$

In practice, we use the approximate variance:

$$g'(\hat{\theta})^2\left(\frac{1}{nI(\hat{\theta})}\right)$$

Exercise: Suppose that $X_1, X_2, \ldots, X_n$ are i.i.d. Exponential(λ). We seek to estimate $\tau = P(X_i > a)$ for some constant $a > 0$. Construct the MLE and find its standard error. Also construct a confidence interval for τ.

By invariance prop., MLE for τ is $e^{-a/\bar{x}}$

(using $\tau = g(\lambda) = e^{-\lambda a}$) $\tau = P(X_i > a) = e^{-\lambda a}$

We know $I(\lambda) = 1/\lambda^2$

$g'(\lambda) = -a e^{-\lambda a}$

So, by Δ-method, can say that $\hat{\tau}$ is approx. $N\left(\tau_0, (g'(\lambda_0))^2\left(\frac{1}{nI(\lambda_0)}\right)\right)$

ie. $N\left(\tau_0, a^2 e^{-2\lambda_0 a}\left(\frac{\lambda_0^2}{n}\right)\right)$

Practical result: $\hat{\pi}$ is approx. normal with variance of $\left(a^2 e^{-2a/\bar{x}}\right)\left(\frac{1}{n\bar{x}^2}\right)$

and SE of $a e^{-a/\bar{x}}\left(\frac{1}{\sqrt{n}\,\bar{x}}\right)$

In general, if $\frac{\hat{\Theta}-\Theta_0}{SE(\hat{\Theta})} \approx N(0,1)$, then $\hat{\Theta} \pm z_{\alpha/2}\, SE(\hat{\Theta})$

is a $100(1-\alpha)\%$ CI for Θ

When does this fail?

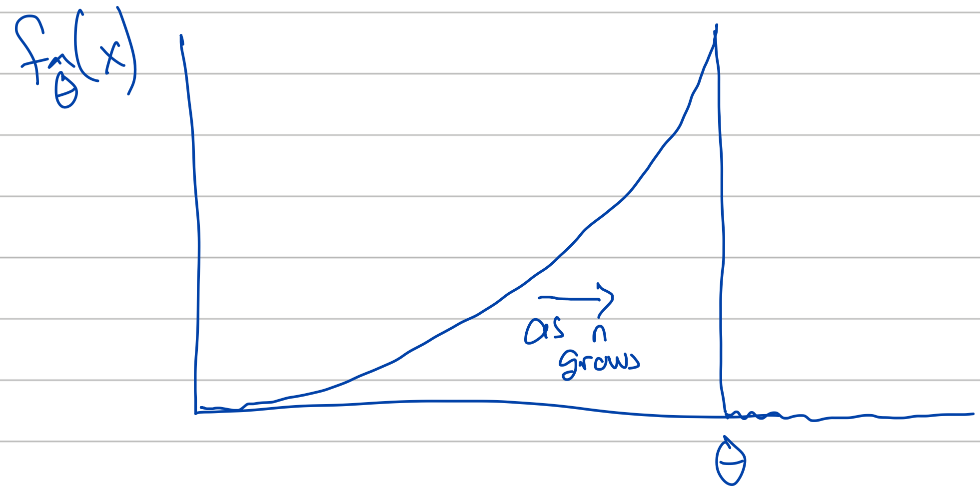
$X_1, X_2, \ldots, X_n$ iid $U(0,\theta)$

What is the dist. for $\hat{\theta}_{MLE}$? Know $\hat{\theta}_{MLE} = X_{(n)}$

$$P(\hat{\theta}_{MLE} \le x) = P(X_{(n)} \le x) = \begin{cases} 0, & x < 0 \\ \left(\frac{x}{\theta}\right)^n, & 0 \le x \le \theta \\ 1, & x > \theta \end{cases}$$

So the density is

$$f_{\hat{\theta}}(x) = \begin{cases} n\left(\frac{x}{\theta}\right)^{n-1}\left(\frac{1}{\theta}\right), & 0 \le x \le \theta \\ 0, & \text{otherwise} \end{cases}$$



Never looks (approx.) normal

Problem: Support of dist. depends on θ, violating one of the "regularity conditions"